

LOGICAL REASONING - CHEAT SHEET

Mid-Term Exam | All Key Formulas & Concepts

1. NUMBERS

Types of Numbers

Natural Numbers: 1, 2, 3, 4, ... (counting numbers, no fractions)

Whole Numbers: 0, 1, 2, 3, ... (natural numbers + zero)

Integers: ..., -3, -2, -1, 0, 1, 2, 3, ... (no fractions/decimals)

Rational: a/b where $b \neq 0$, a & b are integers | **Irrational:** Non-terminating, non-recurring decimals

Even: Divisible by 2 | **Odd:** Not divisible by 2 | **Prime:** Only divisible by 1 and itself

Co-prime: Two numbers whose HCF = 1 (e.g., 14 & 25)

Summation Formulas

Sum of first n natural numbers = $n(n+1)/2$

Sum of first n perfect squares = $n(n+1)(2n+1)/6$

Sum of first n perfect cubes = $[n(n+1)/2]^2$

Sum of first N odd natural numbers = N^2

Sum of first N even natural numbers = $N(N+1)$

Divisibility Rules

Divisor	Rule
2	Last digit is 0, 2, 4, 6, or 8
3	Sum of digits divisible by 3
4	Last two digits divisible by 4
5	Last digit is 0 or 5
6	Divisible by both 2 and 3
8	Last three digits divisible by 8
9	Sum of digits divisible by 9
11	Diff of sum of odd-place & even-place digits is 0 or multiple of 11
12	Divisible by both 3 and 4
25	Last two digits divisible by 25

Factors

If $D = a^p \times b^q \times c^r$, then:

Number of factors = $(p+1)(q+1)(r+1)$

Product of all factors = $D^{(\text{No. of factors} / 2)}$

Key Tips

Primes > 3 are of form $6n \pm 1$ | 2 is the only even prime

Max product of 2 numbers with given sum $S = (S/2)^2$

To check if N is prime: check divisibility by primes up to $\sqrt{N}$

2. PERCENTAGE

Core Formulas

Percentage = (Value / Total) x 100

X% of Y = (X/100) x Y

To convert % to fraction: divide by 100 | Fraction to %: multiply by 100

Increase / Decrease

X% increase: multiply by $(1 + X/100)$ | X% decrease: multiply by $(1 - X/100)$

If A is R% more than B, then B is less than A by: $100 \times R / (100 + R) \%$

If A is R% less than B, then B is more than A by: $100 \times R / (100 - R) \%$

Successive Change

Two successive changes of a% and b%: Net = $a + b + ab/100 \%$

Same % increase then decrease (or vice versa) of R%: Net change = $-R^2/100 \%$ (always a decrease)

Price, Consumption & Expenditure

Expenditure = Price x Consumption

If price increases by R%, consumption must decrease by $100R/(100+R)\%$ to keep expenditure same

Elections / Votes

Valid votes = Total voters - Invalid votes - Non-voters

Winner votes - Loser votes = Majority

3. PROFIT AND LOSS

Basic Definitions

CP = Cost Price | **SP** = Selling Price | **MP** = Marked Price (List Price)

Profit = SP - CP | Loss = CP - SP

Profit % = $(\text{Profit} / \text{CP}) \times 100$ | **Loss %** = $(\text{Loss} / \text{CP}) \times 100$

Discount = MP - SP | **Discount %** = $(\text{Discount} / \text{MP}) \times 100$

Key Relationships

$\text{SP} = \text{CP} \times (100 + \text{Profit}\%) / 100$ | $\text{CP} = \text{SP} \times 100 / (100 + \text{Profit}\%)$

$\text{SP} = \text{MP} \times (100 - \text{Discount}\%) / 100$ | $\text{MP} = \text{SP} \times 100 / (100 - \text{Discount}\%)$

Special Cases

Two items sold at same SP, one at R% profit, other at R% loss: **Net Loss = $R^2/100 \%$**

Two items bought at same CP, one sold at R% profit, other at R% loss: **No Profit No Loss**

Successive Discounts

Two successive discounts of a% and b%: Equivalent single discount = $a + b - ab/100 \%$

Or: Final SP = MP x (1 - a/100) x (1 - b/100)

Dishonest Trader

Claims to sell at CP but uses false weight: **Gain% = $(\text{True Weight} - \text{False Weight}) / \text{False Weight} \times 100$**

*If claims loss of x% but uses false weight: **Gain% = $[(100-x)/100] \times [\text{True Wt}/\text{False Wt}] - 1$ (then x100)***

4. AVERAGE

Core Formula

Average = Sum of observations / Number of observations

Sum = Average x Number of observations

Special Cases

Average of first n natural numbers = $(n+1)/2$

Average of terms in AP (equal difference): **Middle term** (odd count) or **avg of two middle terms**

If each number increases/decreases by k , new average = **old average $\pm k$**

If each number is multiplied by k , new average = **$k \times$ old average**

Assumed Mean Method

Choose an assumed mean (A). Average = $A + (\text{sum of deviations from } A) / n$

Error Correction

If wrong value X was used instead of correct Y :

Correct Average = Wrong Average + $(Y - X) / n$

Addition / Deletion of a Member

When a member is replaced: Change in total = New Average $\times n$ - Old Average $\times n$

New member's value = Old member's value + (Change in avg $\times n$)

Weighted Average

Weighted Avg = $(w_1x_1 + w_2x_2 + \dots + w_nx_n) / (w_1 + w_2 + \dots + w_n)$

5. PERMUTATION AND COMBINATION

Fundamental Counting Principle

Multiplication: If job 1 can be done in m ways AND job 2 in n ways, total = **$m \times n$**

Addition: If job 1 can be done in m ways OR job 2 in n ways, total = **$m + n$**

Permutation (Arrangement - Order Matters)

$nPr = n! / (n-r)! = n(n-1)(n-2)\dots(n-r+1)$

n objects taken all at a time = **$n!$** | Note: $0! = 1$

With repetition: p alike of one kind, q of another: **$n! / (p! \times q! \times r!)$**

Example: MISSISSIPPI = $11! / (4! \times 4! \times 2!)$

Circular Permutation

n distinct objects in a circle = **$(n-1)!$** ways

If clockwise = anticlockwise (necklace/bracelet): **$(n-1)! / 2$** ways

Combination (Selection - Order Doesn't Matter)

$nCr = n! / [r! \times (n-r)!]$

$nCr = nC(n-r)$ | $nC0 = nCn = 1$ | $nC1 = n$

Selecting one or more from n distinct items = **$2^n - 1$**

Key Applications

Handshakes among n people = $nC2 = n(n-1)/2$

Triangles from n points (no 3 collinear) = $nC3$

Parallelograms from M and N parallel lines = $MC2 \times NC2$

6. PROBABILITY

Basic Formula

$P(A) = \text{Favorable outcomes} / \text{Total outcomes} \mid 0 \leq P(A) \leq 1$

$P(\text{not } A) = 1 - P(A) \mid P(A) + P(A') = 1$

$P(\text{certain event}) = 1 \mid P(\text{impossible event}) = 0$

Addition Rule

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

If A & B are mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$

Independent Events

$P(A \text{ and } B) = P(A) \times P(B)$ [if independent]

$P(\text{at least one}) = 1 - P(\text{none})$

Common Scenarios

Coin: $P(H) = P(T) = 1/2 \mid n \text{ coins: Total outcomes} = 2^n$

Dice: $P(\text{any face}) = 1/6 \mid \text{Two dice: Total outcomes} = 36$

Cards: Total = 52, 4 suits x 13 each, 4 Kings, 4 Queens, 4 Jacks

Leap year has 53 of a particular day: $P = 2/7$

7. WORK & CHAIN RULE

Work Formulas

If A does a job in 'a' days, A's 1 day work = $1/a$

A and B together: 1 day work = $1/a + 1/b \mid \text{Time together} = ab / (a+b)$

If A is k times as efficient as B: Time ratio = 1 : k

If A and B together finish in T days, A alone in 'a' days: B alone = $aT / (a - T)$ days

Chain Rule

$M_1 \times D_1 \times H_1 / W_1 = M_2 \times D_2 \times H_2 / W_2$

Where M = Men, D = Days, H = Hours/day, W = Work done

More men -> Less days (inverse) | More work -> More days (direct)

Pipes & Cisterns

Inlet fills in 'a' hrs: Rate = $1/a \mid \text{Outlet empties in 'b' hrs: Rate} = -1/b$

Net rate (both open) = $1/a - 1/b \mid \text{Time to fill} = ab / (b - a)$ [if $b > a$]

8. SERIES - NUMERICAL & ALPHABET

Common Number Series Patterns

Arithmetic: Constant difference (e.g., 2, 5, 8, 11, ... diff = +3)

Geometric: Constant ratio (e.g., 3, 6, 12, 24, ... ratio = x2)

Squares: 1, 4, 9, 16, 25, ... | Cubes: 1, 8, 27, 64, 125, ...

Fibonacci: Each term = sum of previous two (1, 1, 2, 3, 5, 8, 13, ...)

Alternating: Two patterns interleaved | Mixed: +, x, or combination operations

Difference of differences: Check 1st diff, then 2nd diff for pattern

Alphabet Series

A=1, B=2, C=3, ... Z=26 | Use position values to find patterns

Common patterns: Skip letters (+2, +3), reverse order, mirror pairs (A-Z, B-Y)

Opposite letters: A<->Z, B<->Y, C<->X, ... (sum = 27)

9. CODING - DECODING

Common Coding Patterns

Letter Shift: Each letter shifted by fixed positions (e.g., +1: A->B, B->C)

Reverse Coding: Word written in reverse (e.g., CAT -> TAC)

Position Value: Letters replaced by their position numbers (A=1, B=2...)

Symbol/Number Substitution: Letters replaced by symbols or numbers per a rule

Mirror Coding: A<->Z, B<->Y, C<->X (replace with opposite letter)

Approach to Solve

- 1. Compare each letter's position in original and coded word*
- 2. Find the shift/pattern (constant shift, increasing shift, reversal, etc.)*
- 3. Apply the same rule to the question word*
- 4. Check if the pattern involves groups of letters or individual letters*